ALEXANDER NORMAN

(617) 947-5343 | aenorm01@gmail.com | www.linkedin.com/in/alex-norman1 | https://alex-norman.com

SKILLS

Technical: Good Manufacturing Practices (GMP); Computer Aided Design (CAD); Geometric Dimensioning and Tolerancing (GD&T); Six Sigma; Lean; 3D printing; Soldering; Welding; ISO 13485; ISO 14971; 21 Code of Federal Regulations (CFR) 820; Six Sigma

Software/Devices: SolidWorks; Minitab; nTopology; GraphPad; MATLAB; Python; R; C++; Arduino; JavaScript; LaTeX; Preform; Computer Numeric Control (CNC) Machinery; Multisim; Tina-TI; Tinkercad; Oscilloscope; Git; Function Generator; Multimeter; Power Supply; Strain Gauge; Ultrasound; 3D Printer; Analog and digital circuits; breadboard/data acquisition board;; Laser Cutter

CERTIFICATIONS/CERTIFICATES

Certified SolidWorks Professional (SOLIDWORKS); Six Sigma Yellow Belt (American Society of Quality (ASQ)); GD&T Technologist (American Society of Mechanical Engineers (ASME)); ISO 13485 Lead Auditor (ASQ); ISO 14971 Fundamentals (ASQ); GMP (Collaborative Institutional Training Initiative (CITI) Program); 21 CFR 820 Fundamentals (CITI Program); Minitab Fundamentals (Minitab); Medical Device Innovation Fundamentals (Children's Hospital of Philadelphia & University of Pennsylvania)

WORK EXPERIENCE

Philips | Cambridge, MA

Software Test Engineer Intern June - Dec 2026

- Designed and executed system-level validation tests for medical device software integrated with hardware, using data-driven analysis to evaluate performance, reliability, and failure modes
- Applied engineering problem-solving to diagnose complex system issues (software–hardware interactions, network connectivity, edge-case failures), leveraging Python and analytical tools to support root cause analysis
- Collaborated with cross-functional engineering teams to translate test results into design improvements, contributing to iterative development, verification & validation, and overall system optimization

Hebrew University of Jerusalem | Jerusalem, Israel *July 2025 - Feb 2026*

Visiting Scientist

- Developed automated Python-based data processing and visualization tools to analyze experimental datasets, improving efficiency and consistency in interpreting process outcomes
- Designed a MATLAB image-analysis workflow to identify and quantify culture regions of interest (ROI), enabling reproducible measurement and reducing manual analysis variability
- Applied statistical analysis (Kruskal–Wallis testing) to evaluate experimental performance across multiple conditions and identify statistically significant process differences

Wyss Institute at Harvard University | Boston, MA *May - Aug 2023*

Research Assistant Intern

- Used Preform, Insight, and SOLIDWORKS to design 3D printed platforms for microfabricated chips, significantly improving workflow efficiency based on independent observations of prior lab procedures
- Manufactured microfluidic chips to simulate physiological flowrates for testing and development of novel coagulation biosensor
- Conducted Glucose/Fructose/Amylose enzyme assays and analyzed the resulting data using Microsoft Excel to validate the performance of an engineered coating that alters saccharide metabolism
- Corroborated a novel cell culture protocol aimed at optimizing enteroendocrine cell differentiation using immunofluorescence staining

PUBLICATIONS

- Jule, Zesuliwe, et al. "Evolution and Viral Properties of the SARS-CoV-2 BA.3.2 Subvariant." *Virus Evolution*, 2025
- Andrew Seredinski, Tedi Qafko, Nathanael Hillyer, Alexander Norman; Bringing graphene into the undergraduate classroom. *Am. J. Phys.* 1 September 2024; 92 (9): 703–710.
- B. Truong *et al*., "RoboRuggles: Design of a Combat Robot," *2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, Yorktown Heights, NY, USA, 2024, pp. 576-580, doi: 10.1109/UEMCON62879.2024.10754770 (Unaccredited Author)

EDUCATION

Boston University | Boston, MA *September 2024 -May 2026*

Master of Science; Biomedical Engineering

Wentworth Institute of Technology | Boston, MA *August 2021-April 2024*

Bachelor of Science; Biomedical Engineering; Chemistry Minor; Magna Cum Laude ***[GPA 3.82]***

Activities: Wentworth IEEE (E-board); BU IEEE; BU Engineers Without Borders; BU Debate Society; Wentworth Men's Rowing

Relevant Coursework: *Prosthetic Design; Biomechanics; Engineering Mechanics; Introduction to Manufacturing; Fluid Mechanics; Thermodynamics; Solid Mechanics; Biomaterials & Tissue Engineering;* Quantitative Physiology*; Physics I & II; Multivariable Calculus; Differential Equations; Cryptography;* Advanced Circuitry; Bioinstrumentation; Computer Science I